

Day-ahead
forecast

Migration
scheduler

Price each
modeling layer

Decision
rule

Active inter-region migration cuts worst-case emissions **4.0 to 9.9%**; added sophistication (joint covariance, robustness) does not pay until a severity **real grids never reach**.

~0%

value added by joint
covariance and copulas

THE SCREEN

$M^* = 3$

severity where robustness
finally starts to pay

THE PRICE

0 / 17

real grids that reach the
robustness threshold

THE LIMIT

4-10%

worst-case (CVaR) emissions cut
by active inter-region migration

THE LEVER

BACKGROUND

The promise

Modern compute is **flexible**: training and batch jobs shift in time and across data-center sites, so deferring work to **cleaner hours and regions** lowers operational carbon. Across 2021 to 2025 neighbouring grids move strongly together, so spatial structure looks like free signal a robust scheduler could turn into savings. **Question: can measured spatial correlation actually lower out-of-sample carbon?**

DATA & SETUP

What goes in

Electricity Maps hourly carbon intensity, 2021 to 2025 (43,824 obs/zone); train 2021 to 2024, single **locked 2025 test**. Three US and Canadian grids spanning weak to strong residual correlation (up to 0.78). Facility: R zones × 24 h, 50 MW cap, **80% utilization**, 15 MW/h ramp, deferral window 0 to 7 h, inflexible fraction 30 to 75%.

METHOD

How it is built

A Mahalanobis Wasserstein DRO, a second-order cone program in CVXPY (Clarabel/HIGHS), minimizing CVaR of carbon cost (Rockafellar and Uryasev 2000):

$$\min_{x \in \mathcal{X}} (\bar{\rho}, x) + \varepsilon \sqrt{x^\top \hat{\Sigma} x}$$

$(\bar{\rho}, x)$

expected carbon cost:
the dominant mean field

ε

ambiguity radius:
how much we robustify

$\sqrt{x^\top \hat{\Sigma} x}$

robustness penalty: $\hat{\Sigma}$ is the
covariance — correlation
enters ONLY here

Cross-region value enters **only through the off-diagonal blocks of $\hat{\Sigma}$** ; the pre-registered **shuffled-marginals** test zeroes exactly those blocks. We add copula and tail checks, mean ablations, and walk-forward validation to 2025.

RIGOR

Why you can trust a null

194 unit tests, CI on every push

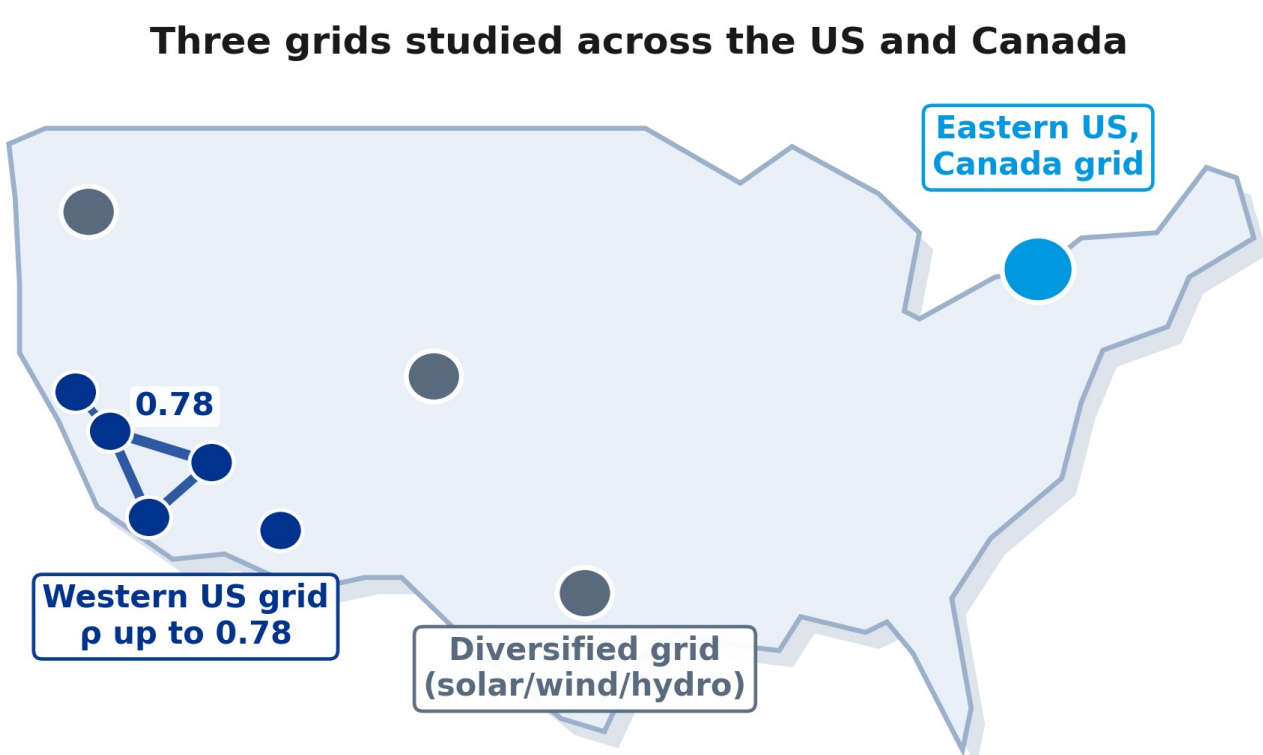
27 cells where TOST confirms a null effect

1 mean-dominance theorem bounding the gap

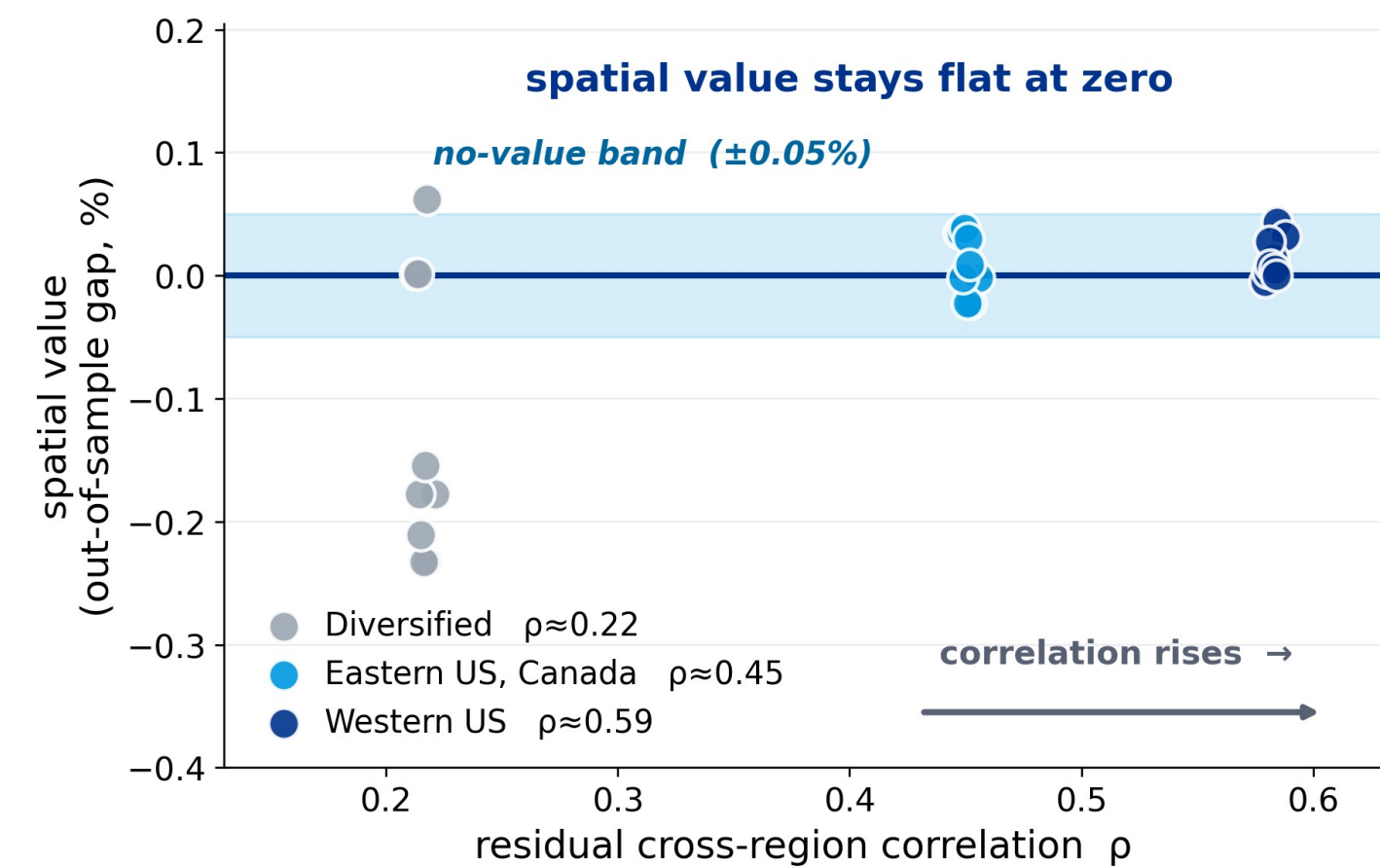
STEP 01 THE PREMISE

The spatial signal is real, but does it pay?

Across 2021 to 2025, de-seasonalized carbon intensity in neighbouring zones moves together. Residual cross-region correlation reaches **0.78** in the Western US and 0.73 in the Eastern interconnection. If spatial structure can ever sharpen a robust schedule, this is the regime where it should.



Three grids span the US and Canada; the **Western US zones form a tightly correlated graph** (up to 0.78).



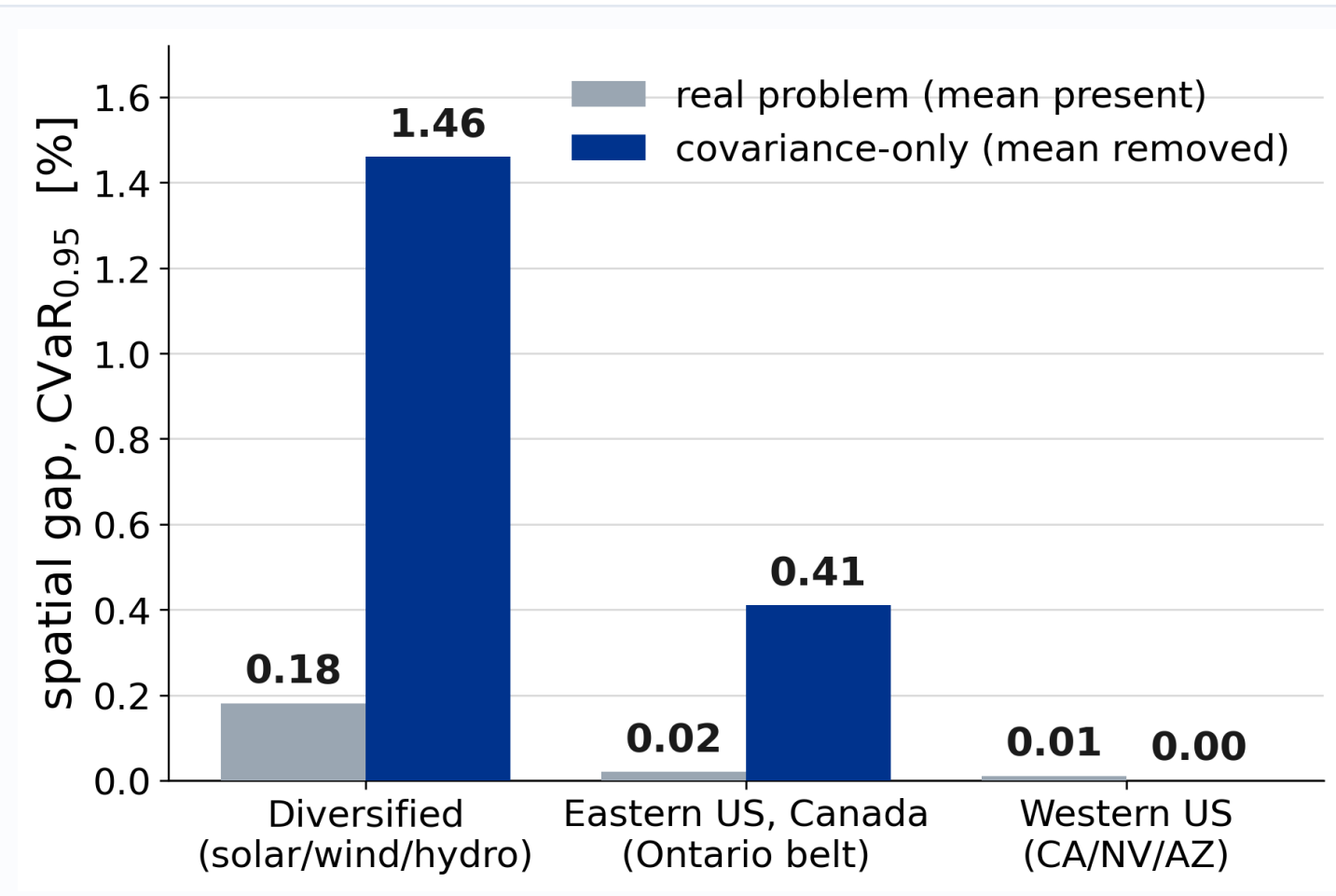
But value does not follow: **out-of-sample spatial value stays at zero** as correlation rises from 0.2 to 0.6.

WE EXPECTED	WE FOUND
Correlation sharpens the schedule	Spatial value stays ~0%
Copulas capture the dependence	Even the maximal copula adds nothing
The signal is simply absent	Real, but masked: +1.46%

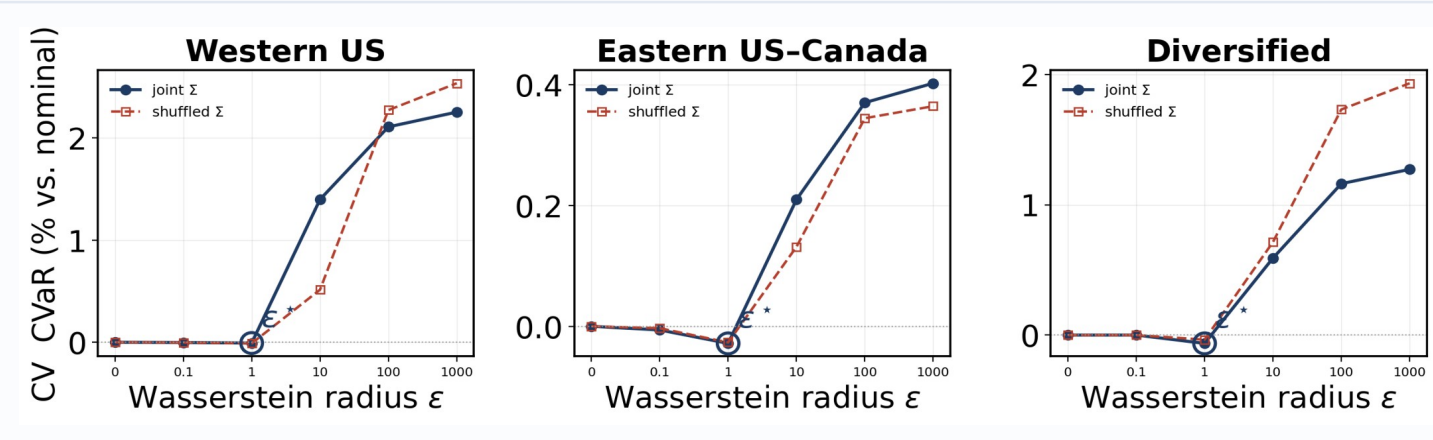
STEP 02 THE SCREEN

Does passive sophistication pay? No.

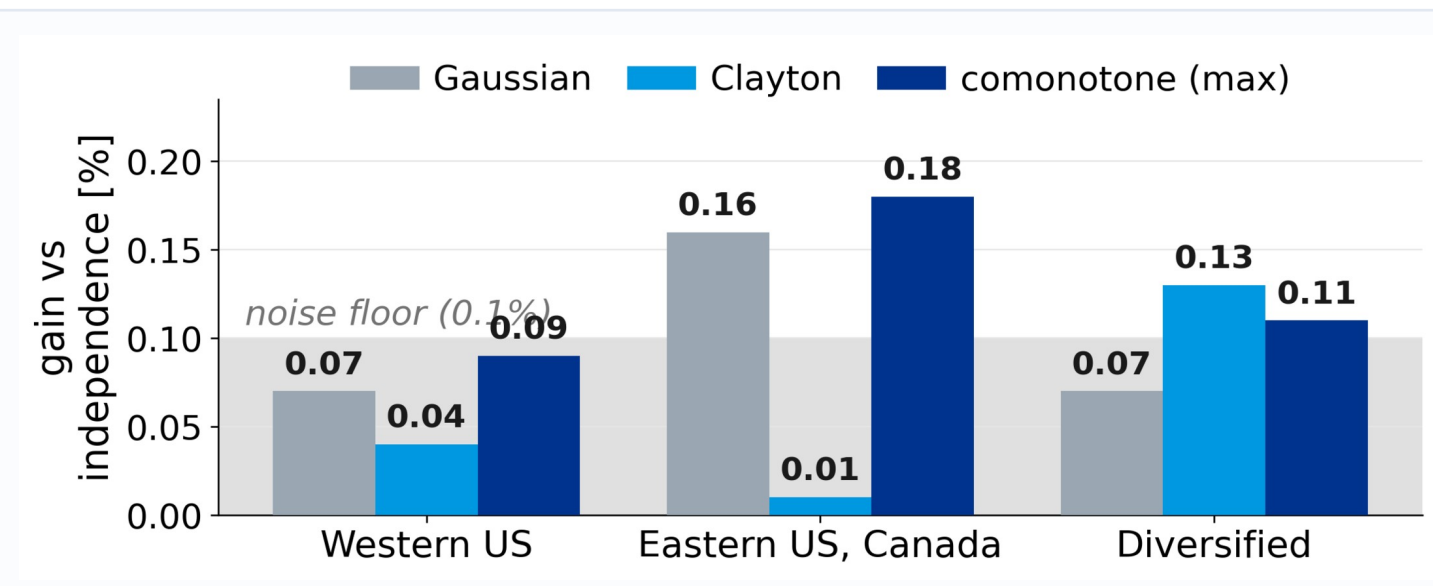
We re-fit the scheduler on the full joint covariance and on a block-diagonal version that erases every cross-region term, then compare out-of-sample CVaR of carbon cost. Across grids and horizons the gap **stays at zero**.



Real problem versus covariance-only: about zero unless the mean field is removed.



Cross-validation picks $\varepsilon^* = 1$: the optimizer truly robustifies, so the null is real.



Gaussian, Clayton, even the comonotone (maximal) copula: **every gain sits at the noise floor**.

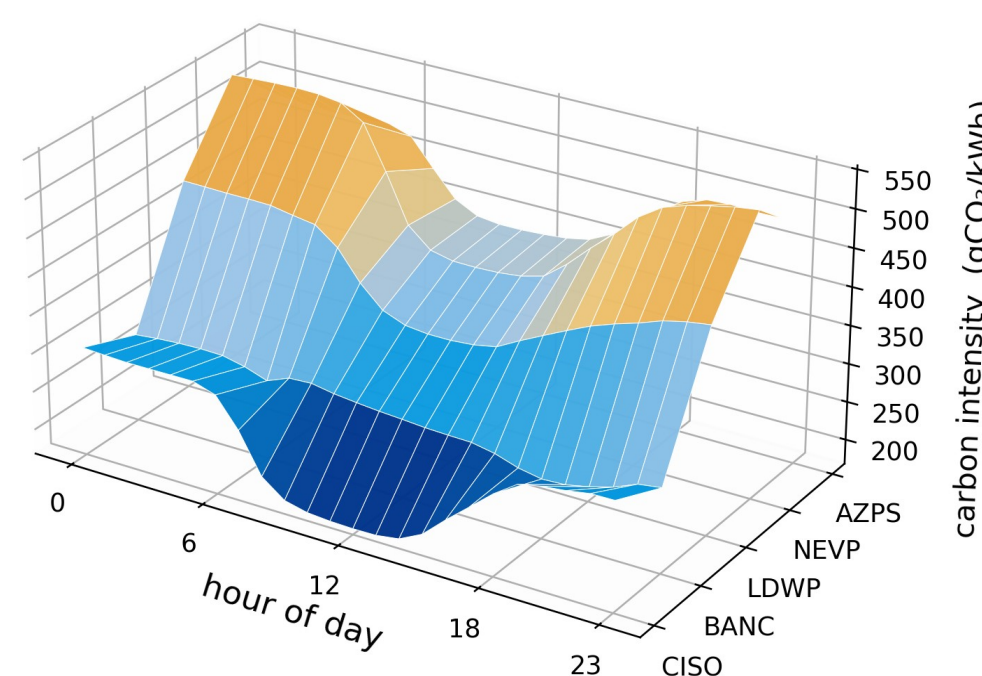
THE VERDICT across every estimator and test, the out-of-sample spatial value is statistically indistinguishable from zero.

STEP 03 THE MECHANISM

Why does it add nothing? The mean field holds it.

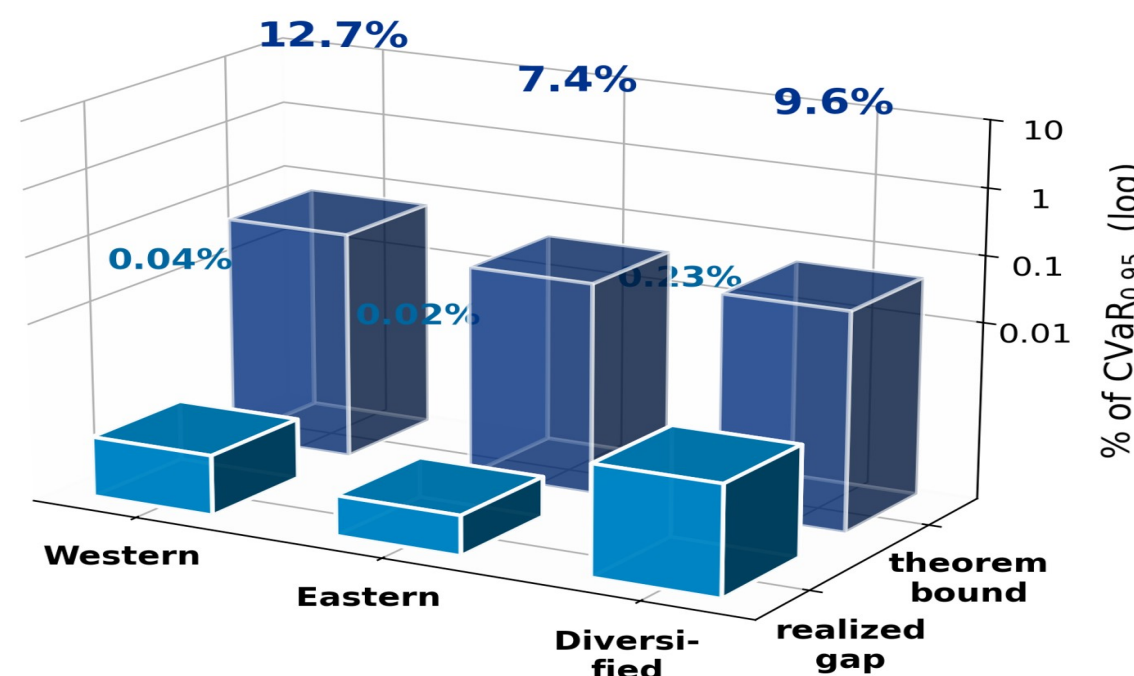
Mean ablation isolates the cause. Hold the covariance fixed and flatten the mean field, and the same spatial structure now pays **+1.46%**. The signal is real but masked by a dominant mean, not absent.

The carbon landscape the scheduler navigates: the dominant mean field



The **dominant mean carbon field**: the schedule follows this terrain; the spatial covariance is a thin ripple on top (≤1.46%).

The theorem bounds the gap — the data beats it by ~100x



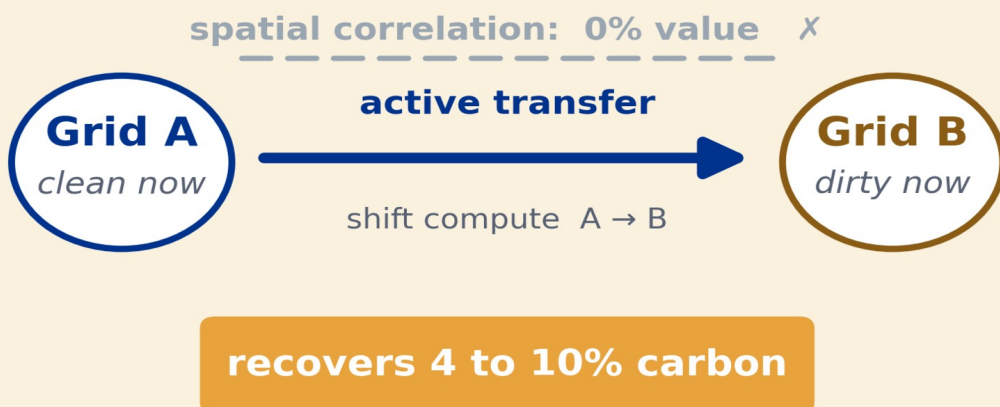
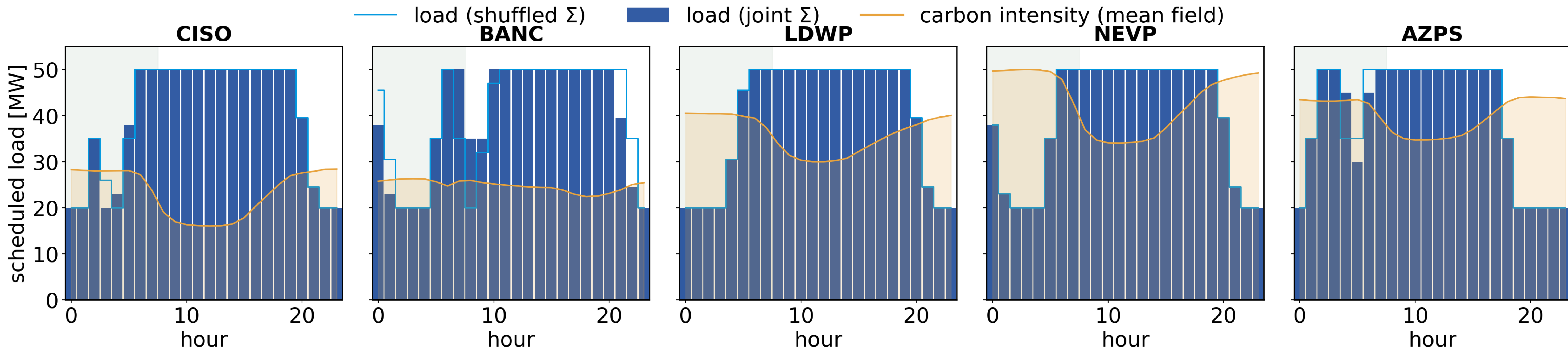
The mean-dominance theorem caps the gap a-priori at **7 to 13% of CVaR**; the realized gap never exceeds 0.23%, about 100x below its own bound.

THE NULL SURVIVES EVERY CHECK

- ✓ Ledoit-Wolf covariance shrinkage
- ✓ Residualization of the mean field
- ✓ Benjamini-Hochberg over 144 cells
- ✓ Walk-forward validation to 2024
- ✓ Ramp and utilization sweeps
- ✓ The full CVaR tail range, 0.80 to 0.99

THE RESULT, SEEN DIRECTLY

Each panel is one Western US grid. Bars: scheduled load. Line: carbon intensity. Joint and shuffled plans overlap.



THE PAYOFF

The value is in physically moving compute toward whichever grid is clean right now. An **active inter-region transfer channel** cuts 4 to 10 percent of carbon, deterministically. The sophisticated layers (joint covariance, copulas, distributional robustness) add nothing until grid emergencies hit 3x severity, which real grids never reach. Deploy the simple migration scheduler; price robustness only if your grid faces genuine extremes.

RECOMMENDATION

Deploy the deterministic migration scheduler; skip joint-covariance and copula modeling; price robustness only past a severity real grids never reach.

REPOSITORY
DATA · THESIS

